

Hawkeye · Pharma EQMS

Engineering Pack · 8-Quarter Execution Plan

Module audit · Architecture gaps · AI/Automation opportunities · Epic breakdown

Vertical: Pharma GMP

Horizon: Q1 2026 → Q4 2027

Generated: 2026-04-20

1 · Current State

Hawkeye today — honest audit

Hawkeye is a production-grade mid-market pharma EQMS with strong domain modelling (150+ models) and solid workflow logic. The core audit-CAPA-deviation flow is fully built. The main risks are in cross-cutting platform primitives: no middleware-enforced multi-tenancy, no observability, no backup/PITR strategy, single-vendor LLM, no SSO/MFA.

Module maturity — at a glance

Module	Status	Headline Gap
Audit Management AUDIT_MANAGEMENT	FULLY BUILT	Internal-audit scheduling only manual
Document Control DOCUMENT_CONTROL	FULLY BUILT	No AI-assisted SOP authoring
CAPA CAPA_MANAGEMENT	FULLY BUILT	Root-cause library not prebuilt
Change Control CHANGE_CONTROL	FUNCTIONAL BUT BASIC	No regulatory-impact auto-assessment
Deviations / Event Mgmt EVENT_MANAGEMENT	FULLY BUILT	No AI-assisted 5-why
Training TRAINING_MANAGEMENT	FULLY BUILT	No role-based curriculum auto-assignment on SOP rev
Risk RISK_MANAGEMENT	FULLY BUILT	No ICH Q9 quantitative Bayesian risk scoring
Supplier Quality SUPPLIER_QUALITY	FUNCTIONAL BUT BASIC	No supplier-audit-to-CAPA closure loop
Management Review MANAGEMENT_REVIEW	FULLY BUILT	No auto-populated KPI dashboard from cross-modules
Asset / Equipment ASSET_MANAGEMENT	FULLY BUILT	No IoT telemetry ingestion (live temp/humidity/vibration)
Chain of Custody CHAIN_OF_CUSTODY	SKELETON ONLY	No automated transfer audit trail
Transaction Review TRANSACTION_REVIEW	SKELETON ONLY	No UI
Regulatory Intel REGULATORY_INTEL	FUNCTIONAL BUT BASIC	FDA-only (no EMA, MHRA, PMDA, CDSCO)
AskHawk AI AI_ASSISTANT	FULLY BUILT	Single LLM vendor (OpenAI)
RFQ / Procurement RFQ_PROCUREMENT	FULLY BUILT	No auditor-rating-to-quote linkage

Cross-cutting capabilities

OK **E-SIGNATURES / 21 CFR 11**

electronicSignatureModel (append-only)
· SHA-256 ·
AUTHORED/REVIEWED/APPROVED
types · covers all major entities.

OK **RBAC**

roleMiddleware + permit() on all routes ·
8 roles · accessGrantModel for resource-
level grants.

OK **NOTIFICATIONS**

notificationPolicyModel + outbox +
preferences · email/SMS/in-app.

BASIC **TENANT ISOLATION**

Field-level `tenantId` required on
every query — no middleware
enforcement. Cross-tenant leak risk.

BASIC **FILE UPLOADS**

evidenceUploadModel + SHA-256 + S3-
compatible · no streaming/virus scan.

BASIC **SEARCH**

Mongo text indexes only · no
Elasticsearch/OpenSearch wiring.

BASIC **REPORTING / PDF**

Audit-report PDF only · no
parameterised report builder.

MISSING **OBSERVABILITY**

No Pino/Winston · no Prometheus · no
OTel · no Sentry. **Blocker for SOC 2.**

MISSING **SSO / MFA**

`requireMFA` flag exists · no
TOTP/WebAuthn · no SAML/OIDC.
Blocker for enterprise.

2 · Architecture Gaps

Top-10 architecture gaps blocking 1000+ tenants + regulatory validation

1. **Multi-tenancy data isolation risk.** No middleware-enforced scoping of queries. Every controller must manually filter by `tenantId`. One forgotten filter = cross-tenant breach. *Fix: Mongoose plugin that auto-injects `tenantId` filter from `req.tenantId`.*
2. **No distributed cache.** In-memory Map cache limits horizontal scaling. *Fix: Redis with tenant-keyed namespaces.*
3. **Backup + PITR undefined.** Hard deletes only · no soft-delete · no documented RTO/RPO. *Fix: immutable delete log + daily Atlas snapshots + documented DR.*
4. **Observability fully absent.** No logs · metrics · traces. *Fix: Pino + Prometheus + OpenTelemetry + Sentry.*
5. **Hash-chaining incomplete.** `contentHashBefore/After` exist but never computed. *Fix: SHA-256 hashing on every CREATE/UPDATE in `dataIntegrityLog`.*
6. **MFA feature-flag incomplete.** No TOTP/SMS. *Fix: WebAuthn + TOTP + SMS fallback. Enforce via `tenant.requireMFA`.*
7. **No outgoing webhooks / async queue.** *Fix: BullMQ on Redis · standard signed-webhook retry queue.*
8. **Single LLM vendor (OpenAI).** *Fix: LLM gateway with pluggable providers (Anthropic · Azure · local). Tenant-level choice for GxP paranoia.*
9. **Admin scopes not enforced at query level.** `requireAdminScope()` is route-only. *Fix: query filter derived from `req.user.adminScope`.*
10. **No IaC / multi-region.** *Fix: Terraform + cross-region MongoDB replica sets + regional object stores.*

Architecture today

Today (Q4 2025)

Gaps

- Multi-tenancy enforced by convention, not middleware
- Hash-chaining fields exist but never computed
- No observability (no Pino/Prometheus/OTel/Sentry)
- No async queue, no outgoing webhooks
- Single-vendor LLM (OpenAI)
- No SSO/MFA implementation (flags only)
- No IaC, no multi-region, no DR

3 · Modules + AI

Module-by-module — gaps + AI/automation opportunities

Audit Management

AUDIT_MANAGEMENT

FULLY BUILT

Current evidence: auditPlanModel, auditRequestsMasterModel, assessmentModel, evidenceModel, full routes + frontend, RFQ-to-award flow

GAPS

- Internal-audit scheduling only manual
- No AI finding-drafting from evidence attachments
- No regulatory-mapping to FDA 21 CFR sections

AI / AUTOMATION OPPORTUNITIES

- LLM drafts observations from uploaded evidence + questionnaire gaps (CAPA auto-suggest)
- Risk-based audit-program auto-scheduler — flag suppliers/departments by deviation density
- Computer-vision on facility walk-through videos for compliance hotspotting

VS COMPETITORS

Veeva + MasterControl + TrackWise all ship AI summarisation.
Hawkeye on par (AskHawk) but lacks evidence-aware finding drafting.

HAWKEYE MOAT

Cross-persona section-assignment + RFQ-to-auditor marketplace is unique; competitors assume in-house audit team only.

Document Control

DOCUMENT_CONTROL

FULLY BUILT

Current evidence: documentControlModel + documentViewModel (redacted views) + full lifecycle routes + frontend

GAPS

- No AI-assisted SOP authoring
- No cross-framework evidence tagging (QMSR vs ISO 13485 vs ISO 9001)
- No structured change-impact analysis
- No Office/Google Docs round-trip

AI / AUTOMATION OPPORTUNITIES

- Draft new SOP rev from a diff of regulatory changes (FDA warning letter → SOP update)
- Auto-tag controlled docs against multiple frameworks (21 CFR, Annex 11, ISO 13485, ISO 9001, IATF 16949)
- Semantic doc search + AskHawk Q&A over controlled-doc corpus with citation

VS COMPETITORS

Qualio Compliance Intelligence (Oct 2025) does multi-framework gap analysis; Veeva AI drafts doc summaries. Hawkeye behind on multi-framework.

HAWKEYE MOAT

Redacted view model (buyer vs auditor) is a differentiator — rare in competitors.

CAPA CAPA_MANAGEMENT

FULLY BUILT

Current evidence: capaModel + capaV2Models + full routes + linkage from audit findings

GAPS

- Root-cause library not prebuilt
- No effectiveness-check statistical validation
- Preventive-action cross-module propagation weak

AI / AUTOMATION OPPORTUNITIES

1. LLM suggests root cause + CAPA plan from deviation narrative (Veeva pilot: 75% cycle reduction)
2. Preventive-action propagation — find similar risks across batches/lines/products, auto-open linked CAPAs
3. Effectiveness-check scheduler with statistical-power calculator

VS COMPETITORS

Veeva AI Agents + AmpleLogic cite 40% faster closure, 30-40% fewer repeats. Hawkeye AskHawk can do this with additional prompting — not yet a dedicated flow.

HAWKEYE MOAT

CAPA→deviation→batch record linkage via cross-module audit trail is a platform edge.

Change Control CHANGE_CONTROL

FUNCTIONAL BUT BASIC

Current evidence: changeControlModel with approvalSteps, riskLevel, impactAssessment — 6 routes — UI list view

GAPS

- No regulatory-impact auto-assessment
- No BOM/spec-diff engine
- No linkage to training assignment after change approval
- UI minimal — no impact-assessment detail view

AI / AUTOMATION OPPORTUNITIES

1. Auto-classify change: notifiable vs. CBE-30 vs. PAS via LLM against FDA/EMA rulesets
2. Spec-diff engine: diff BOM or batch formula between old/new rev — highlight validation triggers
3. Auto-open training assignments when SOP rev is tied to a change

VS COMPETITORS

Veeva tight Vault RIM integration handles regulatory classification. ComplianceQuest has IoT-triggered change. Hawkeye needs both.

HAWKEYE MOAT

N/A — this is a catchup area.

Deviations / Event Mgmt EVENT_MANAGEMENT

FULLY BUILT

Current evidence: DeviationModel with FIVE_WHY, FISHBONE, FAULT_TREE methods + batch disposition + 10 routes

GAPS

- No AI-assisted 5-why
- No trending / signal detection across deviations
- No batch-record auto-link

AI / AUTOMATION OPPORTUNITIES

1. LLM-assisted 5-why and fishbone from deviation narrative — pharma-tuned prompt library
2. Signal detection: cluster similar deviations across batches/products, flag emerging trends to QA
3. Automatic batch-record fetch + OOS/OOT contextualisation

VS COMPETITORS

TrackWise AI auto-summarisation + auto-categorisation is strong here. Veeva Agents score 75% cycle reduction claim.

HAWKEYE MOAT

Investigation methods (5-why, fishbone, fault-tree) modelled as structured data — can auto-populate with LLM.

Training TRAINING_MANAGEMENT

FULLY BUILT

Current evidence: TrainingRecordModel with Assessment, competencyLevel, nextRecurrenceDue + docControlId ref + frontend

GAPS

- No role-based curriculum auto-assignment on SOP rev
- No adaptive-difficulty assessments
- No SCORM/xAPI for external courseware
- No classroom-scheduling UI

AI / AUTOMATION OPPORTUNITIES

1. Auto-generate read-and-understood assignments when an SOP rev is published
2. LLM-drafted knowledge-check questions per SOP (with difficulty scaling)
3. Compliance-gap dashboard: 'who hasn't trained on X rev' with auto-escalation

VS COMPETITORS

ComplianceWire is the training-LMS incumbent (3.6M users). Hawkeye competes on integrated training-tied-to-SOP — structurally better, feature-lighter.

HAWKEYE MOAT

Training tied to SOP revision graph (not just a separate LMS) — integrated approach wins when buyer wants a single platform.

Risk RISK_MANAGEMENT

FULLY BUILT

Current evidence: RiskItemModel with FMEA (Severity × Occurrence × Detectability → RPN), mitigations

GAPS

- No ICH Q9 quantitative Bayesian risk scoring
- No auto-trigger of CAPA on RPN threshold breach
- No risk heatmap dashboard

AI / AUTOMATION OPPORTUNITIES

1. Bayesian risk scoring with prior-incident learning (feed deviation history into risk prior)
2. Auto-CAPA when RPN crosses tenant-defined threshold
3. LLM-generated risk scenarios from SOP + process description (scenario brainstormer)

VS COMPETITORS

ETQ Reliance uses Acerta for predictive risk. Hawkeye has the schema but not the modelling.

HAWKEYE MOAT

ICH Q9 FMEA modelled natively — easy to extend to AI-augmented scoring.

Supplier Quality SUPPLIER_QUALITY

FUNCTIONAL BUT BASIC

Current evidence: supplierProfileModel + supplierRiskMetrics + publicSignal · read-only risk UI · network graph

GAPS

- No supplier-audit-to-CAPA closure loop
- No supplier scorecard trend analysis
- Risk-signal ingestion limited

AI / AUTOMATION OPPORTUNITIES

1. Ingest FDA warning letters, EMA non-compliance, customs import-alerts → auto-update supplier risk score
2. Supplier CAPA portal (supplier fills + signs; buyer reviews) — already partially built
3. GenAI scorecard narrative generation for supplier committee review

VS COMPETITORS

Veeva Vault + ComplianceQuest have mature supplier quality; Hawkeye has the data model but needs closure-loop workflows.

HAWKEYE MOAT

Public-signal ingestion (FDA 483, warning letters) as first-class risk-score input is a differentiator.

Management Review MANAGEMENT_REVIEW

FULLY BUILT

Current evidence: ManagementReviewModel with InputSections + ActionItems + adequacyDecision — ISO 9001:2015 clause 9.3 compliant

GAPS

- No auto-populated KPI dashboard from cross-modules
- No AI exec-summary generator
- No recurring-MRM scheduler

AI / AUTOMATION OPPORTUNITIES

1. Auto-populate MRM inputs: pull CAPA aging, audit findings, deviation trends, training gaps, supplier scorecards — all from one query
2. LLM-drafted exec summary of quarter's quality state (read-ahead for executives)
3. MRM recurring-schedule with Outlook/Google calendar hooks

VS COMPETITORS

Few competitors dedicate a module to MRM. Hawkeye is ahead structurally.

HAWKEYE MOAT

Native MRM module with inputs pulled from cross-module data is a structural advantage over CAPA-centric peers.

Asset / Equipment ASSET_MANAGEMENT

FULLY BUILT

Current evidence: EquipmentModel with CalibrationHistory + statuses + full CRUD · 21 CFR 211.68 tracking

GAPS

- No IoT telemetry ingestion (live temp/humidity/vibration)
- No auto-qualification via IoT readings
- No predictive maintenance

AI / AUTOMATION OPPORTUNITIES

1. IoT ingestion (MQTT, OPC-UA) → auto-log calibration deviations + trigger CAPA
2. Predictive maintenance: ML on run-time + vibration → MTBF prediction, open preventive-action before failure
3. OCR on calibration certificate PDFs + auto-file against equipment

VS COMPETITORS

ComplianceQuest has IoT telemetry; Hawkeye needs it to match.

HAWKEYE MOAT

Equipment → batch-record linkage sets up IoT-to-disposition flow.

Chain of Custody CHAIN_OF_CUSTODY

SKELETON ONLY

Current evidence: workflowSubjectModel with custodyChain array · /coc-tracker page basic

GAPS

- No automated transfer audit trail
- No regulatory metadata per transfer
- No barcode/RFID ingestion
- No temperature-excursion alarms

AI / AUTOMATION OPPORTUNITIES

1. Barcode/QR scan → auto-transfer record with timestamp + handler
2. IoT temp-sensor integration → OOS alarm + CoC break detection
3. LLM incident narrative generator when CoC breaks (drafts a deviation)

VS COMPETITORS

DSCSA-focused vendors (TraceLink, RfXcel) dominate package-level CoC. Hawkeye can position at the internal-process level.

HAWKEYE MOAT

Internal CoC (sample-to-lab, batch-to-warehouse) is under-served by DSCSA vendors — Hawkeye can own this niche.

Transaction Review TRANSACTION_REVIEW

SKELETON ONLY

Current evidence: TransactionReviewModel backend schema · no UI · no SAP/ERP integration

GAPS

- No UI
- No ERP integration
- No high-risk-transaction flagging logic
- No anti-fraud rules engine

AI / AUTOMATION OPPORTUNITIES

1. Anomaly detection on procurement/sale transactions (counterparty risk)
2. LLM-assisted due diligence for high-value transactions
3. ERP push (SAP/NetSuite) webhook ingestion

VS COMPETITORS

Not a standard EQMS module — GRC-adjacent (LogicGate, NAVEX). Hawkeye can skip this or reposition as GRC.

HAWKEYE MOAT

N/A — consider de-scoping or re-positioning as GRC bridge.

Regulatory Intel REGULATORY_INTEL

FUNCTIONAL BUT BASIC

Current evidence: fdalnspectionModel + fda483Model + fdaCitationModel + dashboard · FDA-only

GAPS

- FDA-only (no EMA, MHRA, PMDA, CDSCO)
- No change-control auto-trigger from reg update
- No supplier risk-score integration

AI / AUTOMATION OPPORTUNITIES

1. Multi-agency RSS + AI summarisation (FDA, EMA, MHRA, PMDA, WHO, Health Canada, TGA)
2. Auto-trigger change-control drafts when new regulation affects a tenant's products
3. Warning-letter pattern mining: identify trending FDA concerns for tenant's product class

VS COMPETITORS

Qualio Compliance Intelligence (Oct 2025) covers FDA QMSR + ISO + MDSAP. Hawkeye's FDA dashboard is isolated — needs expansion.

HAWKEYE MOAT

Tenant-specific 'what applies to me' regulatory-feed with CAPA trigger is under-served industry-wide.

AskHawk AI AI_ASSISTANT

FULLY BUILT

Current evidence: Comprehensive RAG stack — 17 routes, conversations, policies, playbooks, unanswered tracking, KB ingest, evals

GAPS

- Single LLM vendor (OpenAI)
- In-memory embedding cache only — no vector DB (Pinecone/Weaviate)
- No tenant-specific KB isolation verified
- No feedback-loop to improve from unanswered

AI / AUTOMATION OPPORTUNITIES

1. Pluggable LLM layer (Anthropic/OpenAI/Azure/local) — GxP/FDA may require on-prem LLM
2. Dedicated vector DB (Pinecone/Weaviate/pgvector) — scale + tenant isolation
3. Active-learning: close the unanswered-question loop into KB articles
4. Grounding guarantees: every answer must cite a doc ID; no ungrounded generation

VS COMPETITORS

Dot Compliance's 'Dottie' is similar positioning. Veeva Agents + TrackWise AI are more mature.

HAWKEYE MOAT

AskHawk's cross-module tool invocation (getAuditSummary, listCapas, getEvidenceList) is a solid differentiator if verified grounded.

RFQ / Procurement RFQ_PROCUREMENT

FULLY BUILT

Current evidence: auditRfqModel + quoteModel + threadModel — full draft → publish → award flow

GAPS

- No auditor-rating-to-quote linkage
- No RFQ templates per audit type
- No auto-matching of auditor specialties

AI / AUTOMATION OPPORTUNITIES

1. Auto-match auditors to RFQ by past audit performance + specialty tags
2. LLM-drafted RFQ from audit-program context
3. Quote-comparison dashboard with AI narrative on trade-offs

VS COMPETITORS

Unique in EQMS — most competitors assume in-house audit team.
Hawkeye's model enables CDMOs to buy-in audits.

HAWKEYE MOAT

Major strategic differentiator — RFQ-to-auditor marketplace has no EQMS competitor.

4 · Target Architecture

Target architecture — 4-phase evolution

Incremental. Each phase is shippable on its own. Later phases depend on earlier phases. Every phase delivers customer-facing value and a platform capability.

Phase 1 · Q1–Q2 2026 — Trust Layer

Delivers

- Integrity proofs for FDA
- Observability for SOC 2
- Multi-LLM + scalable RAG
- Async-first eventing

Phase 2 · Q3–Q4 2026 — Platform Layer

Delivers

- Tenant-authored workflows
- Multi-vertical in one platform
- Enterprise auth story
- Analytics at lake scale

Phase 3 · Q1–Q2 2027 — Validated SaaS + IoT + Agentic AI

Delivers

- Live IoT → CAPA
- Validated SaaS (delivered)
- Agentic AI (acts, not just answers)
- Tamper-proof immutable records

Phase 4 · Q3–Q4 2027 — Global Scale + Adjacencies

Delivers

- Data residency per region
- Partner ecosystem (PLG)
- GRC + Clinical adjacencies
- FedRAMP Moderate in-process

Platform primitives — what we build in what order

PHASE 1 · TRUST LAYER

- Tenant-scope middleware (Mongoose plugin)
- Hash-chain engine (SHA-256 + Merkle root per tenant/day)
- LLM gateway + vector DB (pgvector → Pinecone at scale)
- Observability (Pino · Prometheus · OTel · Sentry)
- BullMQ queue + signed outgoing webhooks

PHASE 2 · PLATFORM LAYER

- No-code workflow engine (visual state-machine editor over existing workflowDefinition)
- Industry-pack registry (pharma · device · ISO 9001 · food · auto)
- Data lake CDC stream (S3 Iceberg or Snowflake)
- SAML + OIDC + WebAuthn; CASL/OPA row-level security

PHASE 3 · VALIDATED + IOT + AGENTIC

- Validation-kit generator (IQ/OQ/PQ/RTM from tenant config)
- IoT ingestion gateway (MQTT · OPC-UA) → equipment + CoC
- AskHawk agentic workflows (actions with e-sig + citations)
- WORM storage + optional blockchain anchor

PHASE 4 · GLOBAL SCALE

- Multi-region (US · EU · APAC) with per-tenant residency
- Terraform for all infra · K8s control plane
- Integration marketplace (SAP · Veeva · LIMS · MES)
- FedRAMP Moderate in-process

"Limitless data collection" — the data spine

Data flow — write path

5 · Industry Verticals

Multi-vertical extension — one platform, many industry packs

Each industry pack is a **config + vocabulary overlay + module-specific schemas** installed on the shared platform. No forking. The platform already supports this pattern via `ModuleConfig.industryProfile` + `vocabularyOverrides` .

Industry-pack anatomy

A vertical pack = **(1)** module on/off matrix · **(2)** regulatory control mapping (ISO clause → module) · **(3)** vocabulary overlay (audit/finding/capa per region) · **(4)** vertical-specific schemas (e.g. HACCP plans for food, APQP for auto) · **(5)** pre-built workflows + templates · **(6)** pre-validated IQ/OQ package (where applicable).

Pharma GMP `PHARMA_GMP`

Primary — live seed (Novex Pharma)

REGULATIONS

21 CFR 210/211 · EU GMP + Annex 11 · ICH Q7 · ICH Q9 · ICH Q10 · WHO GMP · DSCSA

VERTICAL-SPECIFIC MODELS

Deviations, Batch Records, Qualification (IQ/OQ/PQ)

CUSTOMER EXAMPLES

Big Pharma · Generics · CDMOs · Biotech

Medical Device `MEDICAL_DEVICE`

Module skeleton (design-controls, complaint-manager present)

REGULATIONS

ISO 13485:2016 · FDA QMSR (Feb 2026) · MDSAP · EU MDR/IVDR · ISO 14971 (risk)

VERTICAL-SPECIFIC MODELS

Design Controls, Complaints/PMS, Post-Market Surveillance, UDI, IFU/Labels

CUSTOMER EXAMPLES

Class I–III devices · IVD · Combination products

Food Safety `FOOD_SAFETY`

Not built — module flag + industry profile ready

REGULATIONS

FSMA (US) · ISO 22000 · HACCP · SQF · BRCGS · FSSC 22000

VERTICAL-SPECIFIC MODELS

HACCP plans, Allergen management, Recall mgmt, Traceability (farm-to-fork)

CUSTOMER EXAMPLES

Food manufacturing · beverage · ingredients

Automotive AUTOMOTIVE

Not built — module flag ready

REGULATIONS

IATF 16949 · VDA 6.3 · APQP · PPAP · Core Tools
(FMEA/MSA/SPC/Control Plan)

CUSTOMER EXAMPLES

OEMs · Tier 1/2/3 suppliers

VERTICAL-SPECIFIC MODELS

APQP/PPAP packs, Control plans, Layered Process Audits, SPC

Aerospace & Defence AEROSPACE

Not built

REGULATIONS

AS9100D · AS9110C · AS9120B · NADCAP

CUSTOMER EXAMPLES

OEM + MRO + Tier suppliers

VERTICAL-SPECIFIC MODELS

Configuration mgmt, FAI (First Article), Counterfeit-parts mgmt,
Foreign Object Debris

ISO 9001 (generic) ISO_9001

Module config + vocabulary overrides support this

REGULATIONS

ISO 9001:2015 · ISO 45001 (H&S) · ISO 27001 (info security) · ISO
14001 (environment)

CUSTOMER EXAMPLES

Any regulated manufacturing / services

VERTICAL-SPECIFIC MODELS

Context-of-org, Interested-parties register, Objectives tracker,
Integrated-mgmt-system (IMS)

Organic / Agri ORGANIC_FARMING

Industry profile ready

REGULATIONS

USDA NOP · EU 2018/848 · JAS · COR · Fairtrade

CUSTOMER EXAMPLES

Farms · co-ops · ingredients suppliers

VERTICAL-SPECIFIC MODELS

Field records, Input materials, Conversion-period tracking, Certifier
audits

Forestry / CoC FOREST_COC

Industry profile ready

REGULATIONS

FSC Chain-of-Custody · PEFC · SBP

VERTICAL-SPECIFIC MODELS

Species tracking, Volume reconciliation, Input → output yield, Group certification

CUSTOMER EXAMPLES

Forest managers · sawmills · paper mills

Real Estate / AML REAL_ESTATE

TransactionReview module present (backend only)

REGULATIONS

AML / KYC / UBO · FINTRAC (CA) · FinCEN (US) · OFAC · PEP lists · EDD

VERTICAL-SPECIFIC MODELS

Transaction review (already built schema), Beneficial-owner tracking, Suspicious-activity reports

CUSTOMER EXAMPLES

Brokers · title · escrow · developer

High-Ticket / Luxury AML HIGH_TICKET

Not built — uses AML overlay

REGULATIONS

AML for art, precious stones, luxury goods · Basel AML Index · FATF

VERTICAL-SPECIFIC MODELS

Provenance tracking, Appraisal records, Beneficial-owner checks, Sanctions screening

CUSTOMER EXAMPLES

Art dealers · luxury retail · gems/jewellery

Industry-pack ship velocity

Once the vertical-factory engine is live (Q3 2026), each new pack takes 4–6 weeks of domain work + 1–2 weeks of engineering. Target: 2 new packs per quarter in 2027 · 8 industries live by end of 2027.

6 · Roadmap (engineering view)

24-month roadmap · all engineering-relevant initiatives

Color-coded: engineering = pure eng · both = eng + product/board visibility · board = shown for shared context.

Q1 2026	Harden foundations + ship Annex 11 package
Audit-trail hash-chaining + content-hash	BOTH
Complete SHA-256 hash-chain in dataIntegrityLogModel. Every record change emits an immutable link. Output: cryptographic proof of record integrity to FDA inspectors.	
Tenant-scope query middleware	ENGINEERING
Mongoose plugin auto-injects `tenantId` on all queries. Remove the risk of a developer forgetting the filter.	
Observability baseline (Pino + Prometheus + Sentry)	ENGINEERING
Structured logs, metrics scrape, error tracking. Pre-requisite for SOC 2 and ISO 27001.	
Annex 11 2026 compliance package	BOARD
Deliver validation kit + config templates for the new Annex 11 draft reqs (cybersecurity, DR, AI validation). Pre-sell to EU pharma buyers.	
Multi-framework doc-tagging (FDA + ISO 13485 + ISO 9001)	BOTH
Like Qualio Compliance Intelligence. Tag one doc against many regs; dashboard shows coverage per framework.	

Q2 2026

AI-first deviation & CAPA

Deviation / CAPA AI co-pilot

BOTH

LLM drafts RCA (5-why, fishbone) + CAPA plan from deviation narrative. Target: 40% cycle reduction (Veeva pilot parity).

Pluggable LLM layer (Anthropic + OpenAI + on-prem)

ENGINEERING

Swap provider by tenant config. Required for customers that need on-prem LLM (GxP paranoia).

Dedicated vector DB (pgvector or Pinecone)

ENGINEERING

Replace in-memory Map. Enable tenant-scoped KBs, larger corpora, faster retrieval.

AI Agent cost + quality dashboard

BOARD

Per-tenant AI spend + accuracy + confidence distribution. Pricing lever.

Regulatory feed: EMA + MHRA + PMDA

BOTH

Expand from FDA-only. Multi-agency RSS + LLM summariser + tenant-matching.

Q3 2026

Workflow engine + industry packs v1

No-code workflow builder (state-machine editor)

BOTH

Lift the existing workflowDefinition model into a visual editor. Tenants design their own CAPA approval chains without engineering.

Medical Device vertical (QMSR-ready)

BOTH

Design Controls + Complaints/PMS pack. Target FDA QMSR Feb 2026 compliance. First industry-pack template.

ISO 9001 pack + vocabulary overlay

BOARD

Strip pharma-specific language; ship ISO 9001 generic-manufacturing config. Open up the 6-8x larger vertical market.

MFA (TOTP + WebAuthn) + SSO (SAML + OIDC)

ENGINEERING

Wire the existing `requireMFA` flag. Add Okta/Azure AD SAML. Blocker for enterprise sales + SOC 2.

Outgoing webhooks + async job queue (BullMQ)

ENGINEERING

Move from polling cron to async eventing. Unblocks SAP/LIMS/MES push integrations.

Q4 2026

Validation-as-code + data lake

Validation kit generator

BOTH

Auto-generate IQ/OQ/PQ + RTM from tenant config. Cut validation effort from 6 months to 4 weeks (ValGenesis CSA parity).

Data lake staging (S3 Iceberg / Snowflake)

ENGINEERING

Stream every AuditEvent + dataIntegrityLog to a data lake. Enables cross-tenant analytics (opt-in) and large-scale ML training.

Cross-module analytics dashboards

BOTH

KPI dashboard spanning deviations + CAPAs + audits + training + supplier risk. Auto-populates MRM inputs.

Supplier public-signal ingestion (FDA 483/WL + import alerts)

BOTH

Close the supplier-quality loop. Public compliance events auto-update risk score. Differentiator.

Food Safety vertical (FSMA + HACCP)

BOARD

Second industry pack. Demonstrates the platform-of-platforms story.

Q1 2027

Enterprise validation + marketplaces

Validated SaaS (delivered)

BOARD

Published validation package — customers accept as-is; shortens sales cycle by 3-6 months.

Auditor marketplace scale-up

BOARD

Move from 10s to 100s of vetted auditors. Global regions. Specialty tags (biologics, ADCs, combination products).

IoT ingestion (MQTT/OPC-UA) for equipment + CoC

ENGINEERING

Live temp/humidity/vibration streams → auto-log calibration drift + CoC breaks → auto-CAPA.

Automotive vertical (IATF 16949 + APQP/PPAP)

BOARD

Third industry pack. Tier 1/2/3 auto suppliers.

RBAC→ABAC + row-level security (CASL/OpenPolicyAgent)

ENGINEERING

Graduate from role strings to attribute-based policy. Enterprise reqs demand this.

Q2 2027

Predictive quality + AI agents

Predictive deviation signal detection

BOTH

ML on historical deviation streams — flag emerging process drift before it fails. (Acerta/ETQ parity.)

AskHawk agentic workflows

BOTH

AskHawk doesn't just answer — it acts. 'Close these 3 overdue CAPAs' with citations and e-sig requests.

Partner integration marketplace (SAP + Veeva + LIMS + MES)

BOARD

Certified connector program. Turn integration from a 6-month blocker into an app-store install.

GxP + SOC 2 + ISO 27001 certifications

BOARD

Audit certifications in-hand; unlocks enterprise tier. Typically adds 15-20% ACV.

Immutable-ledger anchoring (WORM storage + optional blockchain anchor)

ENGINEERING

Store hash chain in WORM S3 bucket + optional public-chain anchor. Tamper-proof to FDA standards.

Q3 2027

Global scale + vertical roll-up

Multi-region deployment (US / EU / APAC)

ENGINEERING

Data residency compliance. Per-tenant region selection. MongoDB replica sets + regional object stores.

Aerospace + Forestry + Organic packs

BOARD

Industry-pack factory — 3 new verticals shipped in a quarter via pack templates (time-to-market under 6 weeks each once engine is ready).

Terraform / IaC for all infra

ENGINEERING

Full IaC — required for regulated deployments + DR drills.

Customer-admin toolkit (tenant onboarding self-serve)

BOTH

New tenant goes from signup → first CAPA in <10 minutes. Reduces CSM load, enables PLG motion.

Q4 2027

Platform maturity + adjacencies

GRC extension (Risk Register + Controls + Audit 2-in-1)

BOARD

Extend upwards into GRC. Overlaps with LogicGate/AuditBoard but integrated with QMS data. Targets CFO buyer in addition to CQO.

Clinical adjacency (eTMF + CTMS hooks)

BOARD

Light integration with clinical systems. Expands TAM into clinical ops.

FedRAMP Moderate in-process

ENGINEERING

FedRAMP entry. MasterControl Qx Gov parity. Unlocks government pharma + defence med.

AI-gen validation content + SOP authoring at scale

BOTH

LLM-generated SOP first-drafts from regulatory source → editor → approval. Compresses doc-create time 10×.

7 · Tech Debt + Validation

Tech debt inventory + validation burden

Immediate tech debt

1. **Two parallel CAPA implementations** (capaModel + capaV2Models) — pick one; migrate + retire the other.
2. **Hardcoded state-machines alongside generic workflow engine** — statusTrackerModel vs workflowSubjectModel. Consolidate onto the generic engine.
3. **No global search** — Mongo text indexes only. Elastic/OpenSearch layer needed before 100+ tenants with 10M+ docs.
4. **Hard-delete everywhere** — no soft-delete pattern. Required for audit/compliance. Retrofit `deletedAt` + audit-trail tombstones.
5. **ad-hoc console.log** throughout — replace with structured logger (Pino).
6. **In-memory embedding cache** at 4000-item cap — move to pgvector/Pinecone.
7. **Polling cron jobs** — migrate to BullMQ async queue.
8. **No OpenAPI spec for v2 routes** — swagger-jsdoc set up but not all routes annotated.

Validation burden — shrinking from 6 months to 4 weeks

TODAY Customer does full IQ/OQ/PQ. Typical effort: 6 months + \$100K+ consultancy. Re-validate on every release.	Q4 2026 TARGET Validated SaaS package (MasterControl / Qualio parity). Customer performs PQ-only on configured workflows. Typical effort: 4 weeks + \$10-20K config validation.
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Compliance/certification roadmap

- **SOC 2 Type II** — Q2 2026 audit (after observability lands)
- **ISO 27001** — Q4 2026 certification
- **21 CFR Part 11 package** — Q1 2026 (already compliant; package the evidence)
- **EU GMP Annex 11 2026 package** — Q1 2026 (aligned with final rule Jul 2026)
- **HIPAA** — Q2 2027 (for medical device combination products)
- **FedRAMP Moderate** — Q3-Q4 2027 in-process

8 · Acceptance Criteria

Demo milestones — what "done" looks like

Quarterly demo milestones

Q1 2026 Harden foundations + ship Annex 11 package

- **Audit-trail hash-chaining + content-hash** — demo-able: Complete SHA-256 hash-chain in dataIntegrityLogModel.
- **Tenant-scope query middleware** — demo-able: Mongoose plugin auto-injects `tenantId` on all queries.
- **Observability baseline (Pino + Prometheus + Sentry)** — demo-able: Structured logs, metrics scrape, error tracking.
- **Multi-framework doc-tagging (FDA + ISO 13485 + ISO 9001)** — demo-able: Like Qualio Compliance Intelligence.

Q2 2026 AI-first deviation & CAPA

- **Deviation / CAPA AI co-pilot** — demo-able: LLM drafts RCA (5-why, fishbone) + CAPA plan from deviation narrative.
- **Pluggable LLM layer (Anthropic + OpenAI + on-prem)** — demo-able: Swap provider by tenant config.
- **Dedicated vector DB (pgvector or Pinecone)** — demo-able: Replace in-memory Map.
- **Regulatory feed: EMA + MHRA + PMDA** — demo-able: Expand from FDA-only.

Q3 2026 Workflow engine + industry packs v1

- **No-code workflow builder (state-machine editor)** — demo-able: Lift the existing workflowDefinition model into a visual editor.
- **Medical Device vertical (QMSR-ready)** — demo-able: Design Controls + Complaints/PMS pack.
- **MFA (TOTP + WebAuthn) + SSO (SAML + OIDC)** — demo-able: Wire the existing `requireMFA` flag.
- **Outgoing webhooks + async job queue (BullMQ)** — demo-able: Move from polling cron to async eventing.

Q4 2026 Validation-as-code + data lake

- **Validation kit generator** — demo-able: Auto-generate IQ/OQ/PQ + RTM from tenant config.
- **Data lake staging (S3 Iceberg / Snowflake)** — demo-able: Stream every AuditEvent + dataIntegrityLog to a data lake.
- **Cross-module analytics dashboards** — demo-able: KPI dashboard spanning deviations + CAPAs + audits + training + supplier risk.
- **Supplier public-signal ingestion (FDA 483/WL + import alerts)** — demo-able: Close the supplier-quality loop.

Q1 2027 Enterprise validation + marketplaces

- **IoT ingestion (MQTT/OPC-UA) for equipment + CoC** — demo-able: Live temp/humidity/vibration streams → auto-log calibration drift + CoC breaks → auto-CAPA.
- **RBAC→ABAC + row-level security (CASL/OpenPolicyAgent)** — demo-able: Graduate from role strings to attribute-based policy.

Q2 2027

Predictive quality + AI agents

- **Predictive deviation signal detection** — demo-able: ML on historical deviation streams — flag emerging process drift before it fails.
- **AskHawk agentic workflows** — demo-able: AskHawk doesn't just answer — it acts.
- **Immutable-ledger anchoring (WORM storage + optional blockchain anchor)** — demo-able: Store hash chain in WORM S3 bucket + optional public-chain anchor.

Q3 2027

Global scale + vertical roll-up

- **Multi-region deployment (US / EU / APAC)** — demo-able: Data residency compliance.
- **Terraform / IaC for all infra** — demo-able: Full IaC — required for regulated deployments + DR drills.
- **Customer-admin toolkit (tenant onboarding self-serve)** — demo-able: New tenant goes from signup → first CAPA in <10 minutes.

Q4 2027

Platform maturity + adjacencies

- **FedRAMP Moderate in-process** — demo-able: FedRAMP entry.
- **AI-gen validation content + SOP authoring at scale** — demo-able: LLM-generated SOP first-drafts from regulatory source → editor → approval.

Success KPIs (per quarter)

ENGINEERING HEALTH

- Test coverage $\geq 70\%$ on core EQMS modules
- p95 API latency < 300ms
- Zero SEV-1 incidents per quarter
- CI green on main; < 2h lead-time-for-changes

PRODUCT HEALTH

- NPS ≥ 40 among design partners
- Time-to-first-CAPA < 10 min for new tenant
- AI answer grounded-citation rate $\geq 95\%$
- Feature adoption $\geq 60\%$ for each new module 30 days post-launch